

What Is New in Version 2.0

A major update for native VisionEval workflows

Version 2.0 brings the full Workbench workflow to Windows: create and review scenarios, run VisionEval, compare completed results, and build bounded Hypercube studies. The Windows edition runs VisionEval natively rather than through Docker.

Complete Hypercube workflow

Hypercube is a dedicated fifth tab. Build defines up to two bounded parameter axes, Review confirms scope and cases, Run processes only missing or failed cases, Analyze explores the completed matrix, and Export packages selected cases. Windows uses one native VisionEval execution slot, so cases run sequentially. The Run tab shows one compact batch card while individual case logs remain available in Hypercube.

Find All Changes and focused analysis

Find All Changes scans eligible outputs across verified completed Hypercube cases before you choose an individual variable. Independent Year and Aggregation controls, searchable output and scenario rankings, and cached results help identify what merits a closer look. Response Matrix then examines a selected output, with case details and comparison available from its cells.

Clearer scenario and regional workflows

Create separates regional-package development, project setup, single-file editing, Batch Change, and review. Package previews and validation help keep input libraries, model packages, and regional data distinct. Virginia regional data can support MPO or custom region building when matching source inputs are installed; this does not imply a supported statewide model run.

Native runtime setup

Workbench can connect to an existing compatible R, VE_HOME, and VE_RUNTIME installation, or offer the optional current-user setup for R 4.5.3 and VisionEval VE-40-RC7. The setup verifies the runtime before model runs are enabled. Docker is not required, and the application, runtime, and workspace remain separate.

Coordinated runs and recovery

Standard and Hypercube work share a persistent first-in, first-out queue. A workspace lease prevents two backends from controlling the same workspace. Stop controls terminate the active native R process tree before cleaning partial output, and queued work can recover after Workbench restarts.

Comparison, export, and storage

Compare keeps its result selections separate across views, and cached analysis can be rebuilt from authoritative Datastores. Hypercube Export can create separate ZIP packages for the baseline or up to three completed cases, generating temporary output CSVs without retaining full CSV trees for every case. Storage and resource guidance make large studies easier to plan.

Release scope

This overview covers Windows 11 x64. The installer is unsigned, so Windows may show an unknown-publisher warning. Apple Silicon macOS uses a separate Docker-based edition. Study time and storage estimates are planning guidance, not guarantees. Complete statewide Virginia runs are not supported in Version 2.0.

Windows 11 x64 edition | VisionEval VE-40-RC7 | September 2026